

PART B TEMPLATE

Part B-1

#@APP-FORM-HEMSCAPF@#

1. Excellence #@REL-EVA-RE@#

1. *Quality and pertinence of the project's research and innovation objectives (and the extent to which they are ambitious, and go beyond the state of the art)*

Upgrading Remote Sensing for Smarter Farming: The Vision of PLANTDIGISENSE

In the era of smart farming, knowing that a crop is stressed is no longer enough — the real challenge is understanding what and why. Today's monitoring systems can flag stress but struggle to separate abiotic from biotic causes, since both often leave behind the same visual symptoms and overlapping spectral fingerprints. Even powerful commercial tools like Sentinel-2 have their limits: they see the big picture, but they can't distinguish between stress types.

State of the art: In smart farming, the real challenge is not just to monitor crops, but to decode the signals hidden in their physiology. Current platforms — from Sentinel-2 to CubeSats and hyperspectral missions — each offer valuable capabilities, yet none alone can provide the full truth about what and why stresses a plant. Sentinel platforms have already transformed crop monitoring, but they lack spectral resolution. Conversely, hyperspectral imaging (due to its spectral richness) has been manifested as very useful in plant detection of different biochemical and physiological changes, specific to various stress types (Zhang et al., 2024)¹, so platforms such as ENMAP and Prisma are very promising tools, but they have temporal limitations. SmallSats/CubeSats are emerging as very effective for more frequent Earth observation (Rahali et al. 2024)², but still not very widely used in agriculture. Furthermore, there is still big scarcity of diverse, high-quality datasets necessary for robust AI model training for crop stress evaluation (Walsh et al. 2024, Zandi et al. 2025)³. These inherent trade-offs reveal a clear path forward: **no single platform can meet all requirements**. Instead, the integration of **multi-sensor data fusion**, together with **machine learning techniques** is essential for achieving accurate, reliable, and timely plant stress evaluation (see Figure 1.).

Objective. Enhancing plant stress evaluation (abiotic vs. biotic stress differentiation - see Figure 2.) in cereal crops has the aim of reducing pesticide use and improving crop management and decision-making through data-driven approaches. **Supporting EU sustainability** priorities and closing digitalization gaps in EU Widening countries.

Goals. (G1) Harmonize multi-sensor, multi-scale data into pixel-aligned spatio-temporal stacks (analysis-ready cubes) with calibration, BRDF/atmospheric correction and band-pass alignment; (G2) derive physiology-guided features/sensitive indices and learned representations; ground-based validation by plant physiology– pathophysiology specialist; (G3) train and evaluate ML models for crop stress estimation; (G4) translate outputs to practice via farmer-friendly guidance, and FAIR/DOI'd assets.

Ambition & gap beyond the state of the art. Integrating satellites with UAV/UAS and proximal sensing enhances the accuracy of vegetation indices, enabling more precise crop stress monitoring beyond the current state of the art.

Rationale, hypotheses and measurable ambition. PLANTDIGISENSE combines spectral richness with spatial–temporal coverage and UAV- together with expert, physiology-informed labeling. **H1:** multi-sensor fusion with physiology-guided features improves biotic-vs-abiotic discrimination over any single modality. **H2:** protocol-level harmonization plus *leave-one-farming-area-out* using orthodox methods for comparison. Success criteria: accuracy $\geq 85\%$ and Cohen's $\kappa \geq 0.85$ on pre-registered splits, with ablations per modality.

Pertinence and realism. The work aligns with pesticide-reduction and sustainable farming goals and addresses widening-country needs of better adoption of data-driven approaches, as well as **building an interdisciplinary specialist** capable of transforming complex datasets into actionable insights, for both research and practical applications — with a particular focus on **EU Widening**

¹ Zhang, K., Yan, F., & Liu, P. (2024). The application of hyperspectral imaging for wheat biotic and abiotic stress analysis: A review. *Computers and Electronics in Agriculture*, 221, Article 109008. Published June 1, 2024;

² Rahali, L., Praticò, S., Lanucara, S., & Modica, G. (2024). CubeSat constellations: New era for precision agriculture? *Computers and Electronics in Agriculture*, 216, 109764. <https://doi.org/10.1016/j.compag.2024.109764>;

³ Walsh, J. J., Mangina, E., & Negrão, S. (2024). Advancements in Imaging Sensors and AI for Plant Stress Detection: A Systematic Literature Review. *Plant Phenomics*, Volume 6, Article 0153. DOI: 10.34133/plantphenomics.0153; Zandi, A., Hosseini, S., Kashani Zadeh, H., Tavakolian, K., Cho, B.-K., Vasefi, F., Kim, M. S., & Tavakolian, P. (2025). A systematic review of multi-mode analytics for enhanced plant stress evaluation. *Frontiers in Plant Science*, 16, Article 1545025. DOI: 10.3389/fpls.2025.1545025.

countries, where digital agriculture is still emerging. Realism is ensured via deployments on ≥ 3 farms across 1–2 seasons; pre-specified go/no-go rules; cost-sensitive and effective learning for classification; and consensus labeling to reduce noise.

Expected contributions beyond SoA. (i) Curated, DOI-citable datasets, indices, models and reproducible code; (ii) a documented fusion/validation workflow for stress differentiation; and (iii) farmer-oriented guidance that supports earlier, better-targeted interventions.

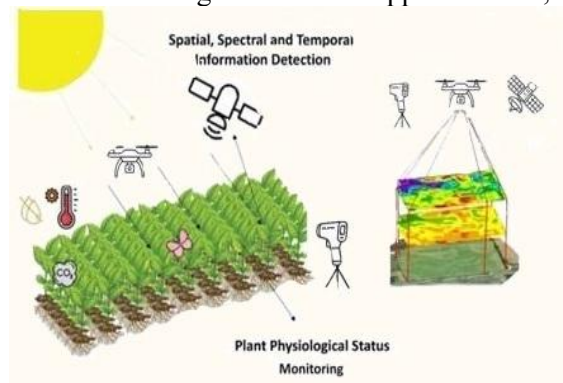


Figure 1. Graphical Abstract of Remote Sensing

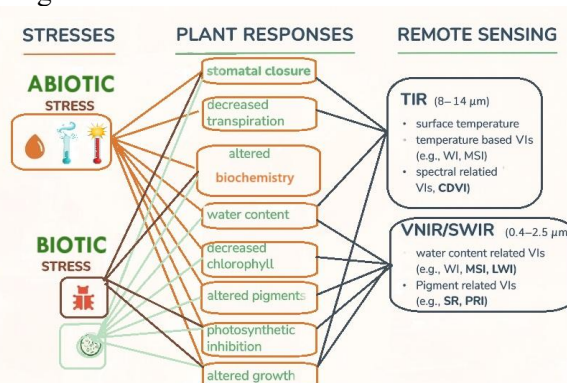


Figure 2. Most important relationships between primary plant stresses, the induced plant responses, and the multi-/hyperspectral TIR remote sensing techniques for the detection of environmental stresses (modified after Gerhards et al., 2019).

Objective	Description	Success measures (verification)	Verifiable outputs / Links
O1. Harmonized multi-sensor data cube & SOPs	Integrate satellite (multispectral - Sentinel-2 & hyperspectral –EnMAP (on availability)), UAV (multispectral & thermal), and proximal hyperspectral data, into analysis-ready cubes with calibration, BRDF/atmospheric correction and band-pass alignment.	SOPs public by M6; co-registration ≤ 0.5 p; $\geq 90\%$ of scenes pass QC; DMP delivered M6 and updated at M12/M24.	Protocols, QC logs, DMP. WP1 · MS1 (M6)
O2. Physiology-guided features & learned representations and validation	Define stress-relevant indices and learned features informed by plant physiology; prepare leak-free split manifests. Validate remote sensing–derived stress detection results with field assessments by the researcher (plant physiology/plant phytopathology licensed).	Feature set documented; index library + benchmarks; ablations show added value; reproducible notebooks. Control groups meet targets (leave one-farm area-out design);	Feature library + notebooks. WP2 → MS2 (M12), MS3 (M18)
O3. ML models & uncertainty for abiotic vs. biotic stress	Train/evaluate RF, SVM, CNN (or equivalent) with calibration metrics and model cards.	Pre-registered splits; accuracy $\geq 85\%$, Cohen’s $\kappa \geq 0.85$, per-class F1 ≥ 0.80 ; calibration (ECE/NLL) reported; model card + code release.	Model repo + card. WP2 → MS3 (M18)

O4. External generalization & translation to practice	Translate outputs to practice via demos, farmer friendly guidance, and FAIR/DOI'd assets;	≥ 2 field demos; practitioner guide post project; ≥ 1 DOI'd release by M24; 24-mo post-release: ≥40 downloads/visits, ≥2 external forks/issues, ≥1 third-party reuse.	Datasets/models with DOIs; demos; guide. WP3 → MS4 (M22), MS5 (M24)
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1.1 ***Soundness of the proposed methodology (including interdisciplinary approaches, consideration of the gender dimension and other diversity aspects if relevant for the research project, and the quality of open science practices).***

Task	Description & Details (compact)
T.1) On-farm setup & monitoring (→ O1)	Cereals (wheat priority); ≥3 farms with diverse management/soils/microclimates; ≥1–2 seasons for robustness. • Farmer co-design (field access, history). Plots marked with RTK-GPS for pixel-level alignment.
T.2) Satellite acquisition (→ O1/O2)	Multispectral: Sentinel-2/Landsat; cadence backup with smallsats. Weekly in peaks; higher during stress; •Hyperspectral:EnMAP (on demand) •standard atmospheric/geo correction + cloud masking.
T.3) UAV acquisition (→ O1/O2)	UAV platform:multispectral / thermal; 2–5 cm GSD; panel calibration pre-flight. • Flights synced to satellite overpasses; altitude/overlap tuned for full-field coverage.
T.4) Ground measurements (→ O2)	Portable hyperspectral + non-destructive physio- pathophysiological analysis: chlorophyll, leaf temperature, stomatal conductance (etc.). • Sampling ≥20 plants/plot at critical phenology & stress events; if needed, collect leaf samples for further assays at cooperating institution.
T.5) Harmonization	Atmospheric/BRDF correction; co-registration to RTK control; bandpass harmonization across sensors → analysis-ready cubes.
T.6) Fusion	Integrate satellite, UAV, ground data into multi-temporal cubes; early/late fusion (e.g., Bayesian, learned encoders). • Output: high-resolution stress maps.
T.7) Features & ML (→ O3)	Indices (e.g., NDVI, PSRI, PRI, WBI, SIPI) + learned features. • Models: RF / SVM / CNN; no data leakage (fixed train/val/test).
T.8) Validation & generalization (→ O4)	Leave-one-farm-area-out control group schemes; Ground validation through visible field symptom estimations by the Researcher (physiology-pathophysiology licensed); stratified splits for class balance. • Metrics: accuracy, sensitivity, specificity, κ ; targets: accuracy ≥ 85%, κ ≥ 0.85; ablations per modality.
T.9) Challenges & mitigation	Clouds/revisit: multi-constellation. •Label scarcity/noise: augmented with semi-supervised/uncertainty-aware learning. •Imbalance: cost-sensitive training + stratified sampling. •Domain shift: harmonization + for each farm leave-one-area-out. •Management confounders: record inputs as covariates. •Licensing/privacy: release derived products; geo-jitter field locations. Covariates (inputs, phenology) are recorded to reduce confounding and go/no-go criteria (accuracy ≥85%; κ ≥0.85) govern iteration.

Concepts, models & assumptions. Our approach assumes that different stress factors can produce characteristic spectral and thermal–physiological responses from leaf to canopy, overcoming the constraint that abiotic and biotic stresses may not always be easily distinguishable. Harmonized

multi-sensor cubes (satellite + UAV + ground) reduce revisit and illumination biases, enabling robust features for ML. Pre-registered splits and leakage-free validation link each step in Table 1.2 to O1–O3, ensuring the method delivers the stated objectives.

Interdisciplinary integration. Plant physiology defines classes and interprets signals; agronomy ensures field feasibility and management relevance; geoinformatics/RS create analysis-ready cubes; ML formalizes discrimination and uncertainty. Integration occurs in three loops: joint sampling/feature design; ablation/error analysis; and co-interpretation of maps with end-users to refine usability.

Gender dimension & diversity in research content. The research analyses plants/fields, not human participants; therefore, sex/gender analysis is not applicable to variables or outcomes. We instead ensure *context diversity* by sampling farms with soils and microclimates. (gender equality in the team will be ensured by the inclusion of 2 female research)

Open science. “As open as possible, as closed as necessary.” We will: (i) publish OA with preprints at submission; (ii) release **anonymised** plot-level data, **processed cubes/index stacks**, model weights and code with DOIs (Zenodo/GitHub); (iii) ensure **reproducibility** via containerised workflows (env files, fixed seeds, split manifests) and run logs; (iv) deliver a **DMP (M6)** with metadata (DCAT/INSPIRE), licensing (docs/code **CC-BY**; data **CC-BY/ODbL** where appropriate), retention, geojittering and removal of farmer identifiers; updates at **M12/M24**.

1.2 *Quality of the supervision, training and of the two-way transfer of knowledge between the researcher and the host*

Aspect	Description
Qualifications and Experience of Supervisors	Prof. András Jung⁴ (ELTE)⁵ — leading expert in hyperspectral remote sensing & environmental modelling; developed cross-device spectral harmonization; co-inventor of a snapshot hyperspectral camera; extensive plant-stress (hyperspectral/thermal) applications; active in COST, Copernicus Ambassador (HU), ESA collaborations. Prof. Gábor Milics⁶ (MATE)⁷ — precision agriculture & tech transfer; co-founder Farmer Digital Agricultural Academy (Digitális Agrárakadémia Plusz); HU representative in AGRITECH EU; role: on-farm monitoring and secondment supervision, farmer-training materials, and network access (farmers/agri-tech). Combined: ≥10 completed PhDs; supervision time/availability confirmed.
Planned Training Activities	Research training: Course 1: Hyperspectral processing, UAV–satellite integration, Geospatial modeling - (ELTE, Institute of Cartography & Geoinformatics). Courses: 2) ML, 3) Statistics, 4) Algorithm Design, 5) RS Processing, 6) Advanced Python - (ELTE Faculties of Science/Informatics). – M18; Online course 7: Horizon Europe Proposal Writing Training - Horizon Europe NCP Portal - M18 ;
Project& Research Management; Horizontal/ Transferable Skills	Research Management: Hands-on proposal writing, project management, reporting, planning; exposure to Horizon Europe structures, deliverables, coordination; participation in COST workshops and digital agricultural advisory; strengthening leadership (task/WP support), interdisciplinary teamwork (scientists–farmers), cross-cultural communication, and outreach/education (farmer-oriented educational materials, methodologies, best practices), Horizontal Transferable skills: acquire ≥3 new skill domains (RS analytics, ML, GIS integration); produce ≥2 interdisciplinary outputs (papers/confs); attend ≥1 international course/summer school; complete ≥1 short secondment/collaboration; PCDP baseline at M1, reviews every 6 months. .
Two-way Transfer of Knowledge	To the researcher (ELTE → fellow): calibration & harmonization SOPs; multi-sensor fusion pipeline reviews (weekly); access to ELTE/ESA/COST networks; on-farm validation guidance. To the host (fellow → ELTE): Improving the host’s capacity in geoinformatics; acquire advanced software/ long-term equipment to support ELTE

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⁵ ELTE - Dr. Jung András; <https://terkep.elte.hu/content/dr-jung-andras.t.15182>

⁶ Dr. Gabor Milics; <https://orcid.org/0000-0002-4499-8322>

⁷ MATE - Dr. Gabor Milics; <https://agronomy.uni-mate.hu/>

Aspect	Description
	capacity and performance, physiology-driven labelling SOP; curated annotation datasets & scripts; EN/HU farmer-oriented training pack, providing access to research data. – Capacity building in EU Widening countries: deliver ≥ 1 farmer-friendly educational methodology; engage ≥ 10 practitioners (farmers/agronomists/policymakers); target $\geq 80\%$ improved understanding (feedback forms). (<i>Facility details are described in Part B2 (Capacity).</i>)
Supervision Arrangements	Schedule: weekly technical meeting (Prof. A.Jung); biweekly applied session (Prof. G. Milics); quarterly milestone reviews; biannual PCDP updates. Back-up supervisor and escalation via doctoral school in place.
Secondment / Short Visits (2–8 weeks total, M12–M24+)	Hosted via MATE – Prof. Gabor Milics, (alternatively, pursued via Digitális Agrárakadémia Plusz) links to an EO/agri-tech partner for product-readiness checks (data ingestion, inference latency, user feedback), stakeholder engagement (non-academic and PhD student farmers) for usability validation and training material co-design. Tasks: Plant monitoring (cereal crops) via satellites, obtaining know-hows on smart farming. Places for the secondments will be farm areas. Expected outputs: UX notes, deployment SOP, and farmer-facing guide draft. Training cadence: weekly technical (ELTE), bi-weekly applied sessions (MATE), quarterly milestone reviews, and biannual PCDP updates; short courses in ML/stats/RS/GIS scheduled M3–M24 with completion certificates logged.

1.4 Quality and appropriateness of the researcher's professional experience, competences and skills

Early-career plant physiologist (PhD in Plant Physiology; 5 years post-PhD), former Assistant Professor at the Agricultural University of Plovdiv; currently researcher at MVRCI (Agricultural Academy, Bulgaria). Expertise: plant stress physiology and non-destructive measurement; experimental design and field trials; UAV remote sensing (licensed pilot); statistics and RS/ML. Education & outputs: PhD (Plant Physiology); BSc/MSc Plant Protection (Pathophysiology); MSc Agricultural Engineering in progress (GIS, Environmental Engineering - expected graduate: May 2026). Publications: 5 peer-reviewed; 2 manuscripts under review on remote sensing of abiotic stress. Networks & projects: MVRCI researcher (since 2024) with AgroHub.BG / AgroDigiRise⁸ EDIH EU links, active in COST Actions⁹ PANGEOS & RECROP and national RS-for-agriculture projects. Skills & upskilling: field experiment design, plant protection advisory service for farmers, ground/UAV data collection, geospatial/physiological analytics, statistics, Python/Data Science/ML (ongoing to Dec 2025). This interdisciplinary profile (physiology–protection–agronomy–data science), combined with practical EDIH engagement, is directly aligned with PLANTDIGISENSE and supports high-quality, farmer-ready outcomes.

2. Impact

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2.1 Credibility of the measures to enhance the career perspectives and employability of the researcher and contribution to his/her skills development

Contribution
The fellowship will equip the researcher with cutting-edge skills in remote sensing, hyperspectral/multispectral data fusion, UAV/UAS, and machine learning for precision plant stress monitoring. Enhancing the Fellow project management skills, interdisciplinary collaboration and scientific communication. It will also prepare the researcher to mentor and tutor farmers through AgroDigiRise EDIH Hubs, fostering digital agriculture literacy. These experiences will position the researcher to pursue a career role at JRC ¹⁰ EU (Ispra), contributing to EU-level precision agriculture

⁸ AgroHub.BG | Where agriculture meets technology (<https://edih.agrohub.bg/>); AgroDigiRise | European Digital Innovation Hubs Network (<https://european-digital-innovation-hubs.ec.europa.eu/edih-catalogue/agrodigirise>)

⁹ COST | European Cooperation in Science and Technology (<https://www.cost.eu/>)

¹⁰ Joint Research Centre; https://commission.europa.eu/about/departments-and-executive-agencies/joint-research-centre_en

Contribution
and environmental monitoring projects, while simultaneously strengthening transferable skills and employability across both academic and non-academic sectors. Also will prepare the Fellow for PI-level roles through mentoring, proposal and manuscript writing.

2.2 Suitability and quality of the measures to maximise expected outcomes and impacts, as set out in the dissemination and exploitation plan, including communication activities^{#@COM-DIS-VIS-CDV@#}

Channel	Audience	Actions & timeline (M-by-when)	Expected KPIs
Journals & conferences	RS/agronomy/ML researchers	Submit ≥ 1 authoring Q1/Q2 paper (M22), preprints at submission; ≥ 1 conference contributions (abstracts/talks/posters) via ELTE, COST/ESA networks (M12–M24).	≥ 1 papers; ≥ 1 conference outputs; preprints online;
Open assets (FAIR)	Academia, RTOs, SMEs (agri-tech)	Release dataset + spectral library + code + model weights with DOIs (M24); maintain versioned repos; DMP by (M6) and updated.	≥ 1 DOI'd releases; ≥ 40 downloads/visits; ≥ 2 external forks/issues; ≥ 1 re-cited.
Stakeholder events & training	During project: On-farm observations and interactive training of involved farmers. Post-project: Knowledge transfer through the AgroHub.BG	≥ 1 farmer friendly training paper (M24).	≥ 30 attendees; $\geq 80\%$ positive feedback, ≥ 2 on-farm pilots/continued
Policy & public communication	Ministries/extension; general public	1 policy brief on remote plant stress evaluation targeting M24; targeted briefings with extension (e.g Farmer Expo), (M18–M24); web/news/social updates per milestone.	≥ 100 brief downloads; ≥ 2 policy/extension meetings; ≥ 2 media mentions;

2.3. The magnitude and importance of the project's contribution to the expected scientific, societal and economic impacts

Impact Type	PLANTDIGISENSE Contribution	Magnitude & Importance
Scientific	First open-access Eastern Europe spectral-physiological crop dataset under real farming conditions. - Multi-platform data fusion (drone, proximal, field sensors) with standardized protocols. - ML workflows for stress detection, cross-validated with physiological assessment.	Benefits hundreds of researchers, enabling robust AI/ML model development and cross-project adoption. Advances precision agriculture and plant physiology knowledge. Added value for PhD students through shared data.
Economic / Technological	Accurate stress differentiation reduces pesticide and input costs. - High-quality datasets improve ML model accuracy (+3% biotic stress detection, +2.9% overall). - Early stress detection prevents yield loss (~2–3%), boosting sector revenue. - Enhances market competitiveness and	Direct cost savings for farmers; improved efficiency and yields; scalable technology adoption in EU Widening Countries.

Impact Type	PLANTDIGISENSE Contribution	Magnitude & Importance
	quality in cereals (covering ~50% of Bulgaria's farmland).	
Societal	On-field transfer via demo sites and stakeholder engagement (post project). Promotes sustainable farming, reduced chemical inputs, and resource efficiency. Supports European Green Deal strategies. Boosts digital capacity in EU Widening Countries. Enhances societal impact by directly transferring knowledge to local communities and promoting the adoption of sustainable digital farming practices.	Broad impact across farmers, students, and agronomists; contributes to climate resilience, sustainability, and faster adoption of precision agriculture tools in underdeveloped regions.

Quantified estimates. As per Section 2.2, we expect ≥ 1 DOI'd dataset/model release by **M24**, reaching ≥ 40 downloads/visits and ≥ 2 external forks/issues within 24 months, with ≥ 1 documented third-party reuse. Technically, we target **accuracy $\geq 85\%$** and **$\kappa \geq 0.85$** on pre-specified leave-farm-area-out design for each farm. Economically, preventing a conservative **2–3%** yield loss on **10,000 ha** during pilots would already outweigh project costs; precise figures will be reported in Part B-2 with regional baselines. Adoption is supported by ≥ 1 demo points and ≥ 1 continued pilots (project) via AgroDigiRise/AgroHub.BG (post project).

PLANTDIGISENSE Go-to-Market – Plugin Approach for Existing Software

Integration: Plugin for Pix4Dfields, QGIS, FieldView, providing stress differentiation and prescription maps without a full standalone system. **Free/Open Version:** Core functionality for farmers, researchers, and students, supporting training and open science. **Premium Tier:** Advanced features and batch processing for B2B users (cooperatives, OEMs). **Follow-up Project:** Planned collaboration to expand plugin capabilities, integrate additional crops/sensors, and broaden adoption. **Broader use:** for open-license software (QGIS).

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3. Quality and Efficiency of the Implementation#@QUA-LIT-QL@# #@WRK-PLA-WP@# #@CON-SOR-CS@# #@PRJ-MGT-PM@#

3.1 *Quality and effectiveness of the work plan, assessment of risks and appropriateness of the effort assigned to work packages*

WP	Task Description	Deliverable(s)	End Month
WP1 Multi-Sensor Data Collection	T.1.1. Develop standardized multi-sensor data collection protocols (RTK-GPS-based UAV flights, hyperspectral calibration, thermal and RGB acquisition)	D.1.1. Standardized multi-sensor data collection protocol	M2
	T.1.2. Conduct first-season field measurements	D.1.2. First-season validated dataset (preliminary, labelled)	M13
	T.1.3. Conduct second-season measurements and integrate with first-season data	D.1.3. Final validated dataset (two seasons, fully labelled)	M22
WP2 Training & Capacity Building	T.2.1. Attend short-term courses and workshops in machine learning, data science, remote sensing, and GIS	D.2.1. Certificates and skill report	M18
	T.2.2. Mentoring early-stage researchers	D.2.2. Reports on capacity	M24

WP	Task Description	Deliverable(s)	End Month
	and student involvement in field/lab work	building	
	T.2.3. Develop in-house expertise in hyperspectral and thermal data preprocessing, vegetation index calculation, and AI model training	D.2.3. Internal training documentation	M18
WP3 Data Analysis & AI Modeling	T.3.1. Preprocess UAV, hyperspectral, and thermal datasets (georeferencing, calibration, cleaning)	D.3.1. Index stack and QC report	M22
	T.3.2. Compute and validate stress-sensitive indexes (NDVI, WBI, PSRI, SIPI, etc.)	D.3.2. Validated stress-specific spectral indices	M22
	T.3.3. Develop and validate ML models for automated stress detection and yield prediction ($\geq 85\%$ accuracy)	D.3.3 Trained and validated ML classification models	M23
	T.3.4. Statistical analysis and visualization (ANOVA, PCA, correlation, regression)	D.3.4 Statistical analysis & visualization compendium	M23
WP4 Dissemination, Communication & Stakeholder Engagement	T.4.1. Present results at international conferences	D.4.1. Conference presentations	M24
	T.4.2. Publish open-access peer-reviewed article(s)	D.4.2. Submission of peer-reviewed open-access publication	M24
	T.4.3. Develop farmer-oriented training materials	D.4.3. Farmer-oriented training materials & dissemination package	M24
	T.4.4. Maintain online presence, share datasets/code on GitHub/Zenodo, participate in MSCA events, and communicate with policymakers	D.4.4 Sustained online presence & repository maintenance log	M24
WP5 Project Management & Coordination	T.5.1. Regular project meetings (weekly/bi-weekly) with ELTE and MATE teams	D.5.1. – Project Management & Quality/Risk Plan (report)	M24
	T.5.2. Financial and administrative management	D.5.2. Mid-term Progress & Risk Update (report)	M24
	T.5.3. Monitoring and risk management	D.5.3. Final Management & Risk Report (report)	M24
Effort (person-months, Fellow)	WP1 Data Collection 8 PM (4 per each season) ; WP2 Training/Capacity 3 PM ; WP3 Analysis & AI 8 PM ; WP4 Dissemination/Stakeholders 3 PM ; WP5 Management 2 PM . Total: 24 PM. <i>Rationale:</i> Front-loads WP1/3 for seasonality and model work; reserves explicit time for training and the 2–8 week secondment within WP2; keeps dissemination active yet proportionate; ensures management is covered.		
Risk Assessment	Likelihood	Impact	Mitigation Strategy
Weather-related delays in field data collection	Medium	Medium	Use of staggered sowing dates, backup field plots, and data from multiple locations.

Risk Assessment	Likelihood	Impact	Mitigation Strategy
Weak stress signal differentiation	Medium	High	Combine spectral data with visual scoring and physiological indicators conducted by the Researcher.
Model underperformance or poor classification accuracy	Medium	Medium	Apply multiple machine learning approaches; adjust model parameters based on feedback.
Administrative or ethical delays at host institution	Low	Low	Early coordination and regular communication with administrative departments.
Delays in equipment availability or malfunction (e.g. sensors)	Low	Medium	Maintain redundant equipment; early procurement and system checks.
Satellite acquisition constraints	Low	High	Define flexible acquisition windows; use multi-constellation backups (Sentinel2/Landsat/smallsats); prioritize scenes via cloud thresholds; fall back to archival imagery; increase UAV sampling during cloudy periods; include 1–2 month schedule buffer.
UAV/satellite, field permissions & access NOTAMs/weather limits; landowner consent delays or withdrawal	Low	Medium	Licensed drone usage by certified pilot (Researcher owns EU drone pilot license); Maintain at least one approved backup field per farm; prepare alternative flight dates and indoor calibration tasks for no-fly days.

3.2 Quality and capacity of the host institutions and participating organizations, including hosting arrangements

Aspect	Details
Hosting Institution	Eötvös Loránd University (ELTE) –at the Faculty of Informatics, specifically within the Institute of Cartography and Geoinformatics Supervisor: Prof. András Jung, expert in remote sensing, geospatial technologies, environmental monitoring.
Hosting Arrangements	Dedicated office/laboratory space at ELTE. Full integration into Department's research groups, weekly meetings, and interdisciplinary workshops.- Administrative and technical support for relocation, project management, procurement, and dissemination.- Access to short-term training courses. Participation in international networks (e.g., COST Action, FAO/EU webinars).
Institutional Infrastructure	GIS and Remote Sensing Laboratory with: • DJI Matrice 350 RTK – Industrial drone with RTK GPS. • DJI Zenmuse L1 – LiDAR + RGB camera. • DJI Zenmuse P1 – 45 MP photogrammetry camera. • DJI Zenmuse H20T – Multi-sensor (thermal, zoom, wide). • Ultris S5 Hyperspectral Camera – VIS–NIR narrow bands. • Cubert CUVIS Lab Lite Package – Hyperspectral imaging with calibration tools.
Software & Data Processing Facilities	Licensed software: ArcGIS, ENVI, MATLAB.- Access to cloud/open-source platforms: Google Earth Engine, QGIS, SNAP, Python, R. High-performance computing clusters for AI and large-scale geospatial analysis. Secure storage and backup for large remote sensing datasets.
Additional Facilities	Field monitoring areas in collaboration with partner farms. Access to ELTE's institutional library, IT services, and open science repositories.

Aspect	Details
Key Participating Organisation	Hungarian University of Agriculture and Life Sciences (MATE) – Collaboration with Prof. Gábor Milics as scientific advisor. Strengthen methodological robustness. Connect with farmers. Co-develop practical farmer guidelines. Facilitate knowledge exchange between academia and practice.
Other Collaborations	1–2 PhD students (also practicing farmers) involved in precision farming projects. Partner farmers, including smallholders, providing monitored fields. Colleagues from MATE Plant Physiology Department for additional physiology measurements.
Integration & Capacity Building	Active collaboration between ELTE, MATE, and field stakeholders. - Development of AI-driven plant stress evaluation methods.- Creation of diverse, high-quality datasets for AI training. - Mentoring of early-stage researchers. - Direct application of results to climate-smart and sustainable farming practices.

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Milestones	
MS1 (M12)	<i>Completion of first season data collection & quality check. <u>Links: O1, O2.</u></i> Verification: Deployment logs, calibration records, preliminary datasets.
MS2 (M22)	<i>Expert labelling & dataset validation. <u>Links: O2.</u></i> Verification: Standardized scoring protocols, annotated datasets, photo evidence
MS3 (M23)	<i>Model performance benchmark achieved (RF/SVM/CNN). <u>Links: O3.</u></i> Verification: Model evaluation report, reproducible code, $\geq 85\%$ accuracy and $\kappa \geq 0.85$.
MS4 (M24)	<i>Practitioner guide draft delivered (paper submitted; ≥ 1 conference presentation). <u>Links: O1–O4.</u></i> Verification: Submission proof, conference acceptance/agenda, demo materials.
MS5 (M24)	<i>Final dissemination & open release. <u>Links: O4.</u></i> Verification: Public DOIs (Zenodo/GitHub) for datasets/models/code, practitioner guide (EN/HU), final reporting package to EC.

Gantt chart PLANTDIGISENSE

